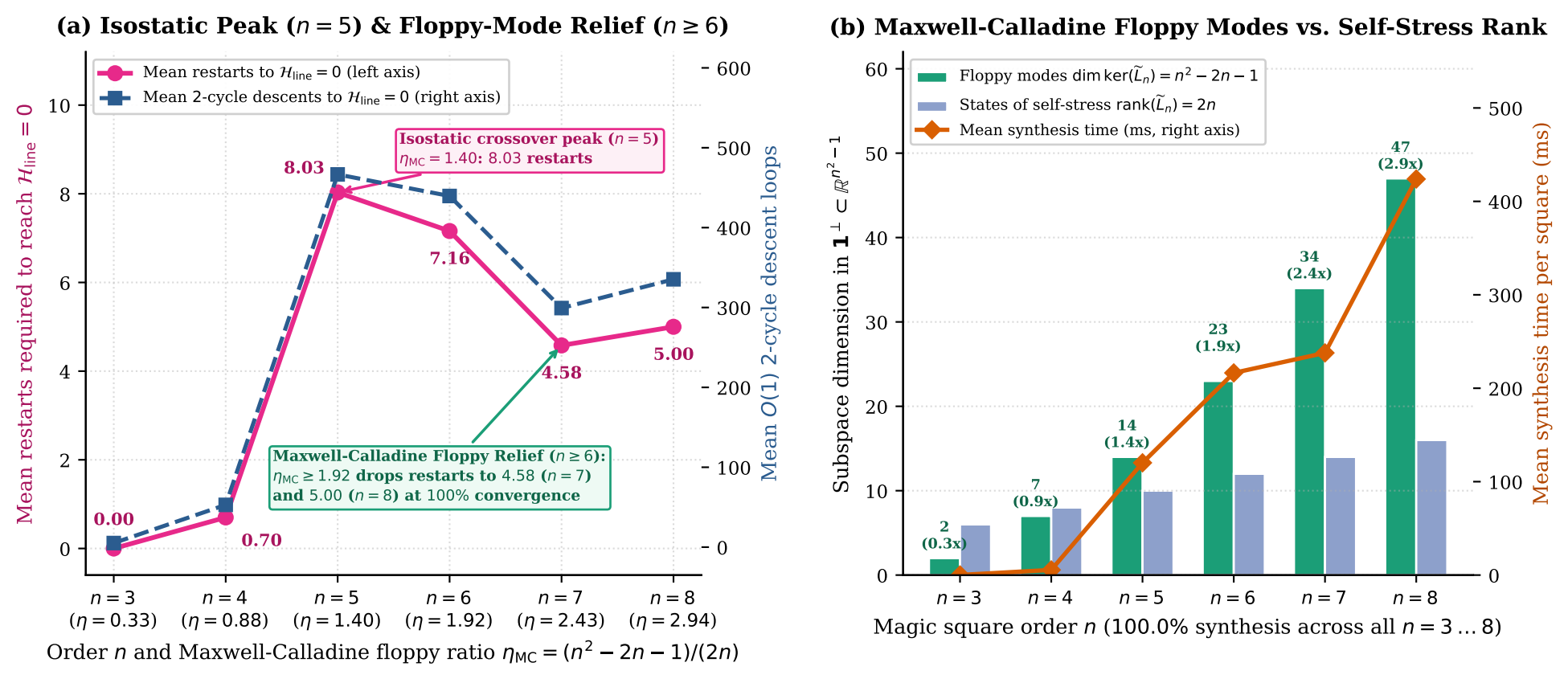

(a) Isostatic Peak ($n = 5$) & Floppy-Mode Relief ($n \geq 6$)
Mean restarts to $\mathcal{H}_{\text{line}} = 0$ (left axis)
Mean 2-cycle descents to $\mathcal{H}_{\text{line}} = 0$ (right axis)
Isostatic crossover peak ($n = 5$)
$\eta_{\text{MC}} = 1.40$: 8.03 restarts
8.03
7.16
4.58
5.00
0.00
0.70
Maxwell-Calladine Floppy Relief ($n \geq 6$):
$\eta_{\text{MC}} \geq 1.92$ drops restarts to 4.58 ($n = 7$)
and 5.00 ($n = 8$) at 100% convergence
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean $O(1)$ 2-cycle descent loops
$n = 3$ ($\eta = 0.33$)
$n = 4$ ($\eta = 0.88$)
$n = 5$ ($\eta = 1.40$)
$n = 6$ ($\eta = 1.92$)
$n = 7$ ($\eta = 2.43$)
$n = 8$ ($\eta = 2.94$)
Order n and Maxwell-Calladine floppy ratio $\eta_{\text{MC}} = (n^2 - 2n - 1)/(2n)$
(b) Maxwell-Calladine Floppy Modes vs. Self-Stress Rank
Floppy modes $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$
States of self-stress $\operatorname{rank}(\widetilde{L}_n) = 2n$
Mean synthesis time (ms, right axis)
2 (0.3x)
7 (0.9x)
14 (1.4x)
23 (1.9x)
34 (2.4x)
47 (2.9x)
Subspace dimension in $\mathbf{1}^{\perp} \subset \mathbb{R}^{n^2-1}$
Mean synthesis time per square (ms)
Magic square order n (100.0% synthesis across all $n = 3 \ldots 8$)